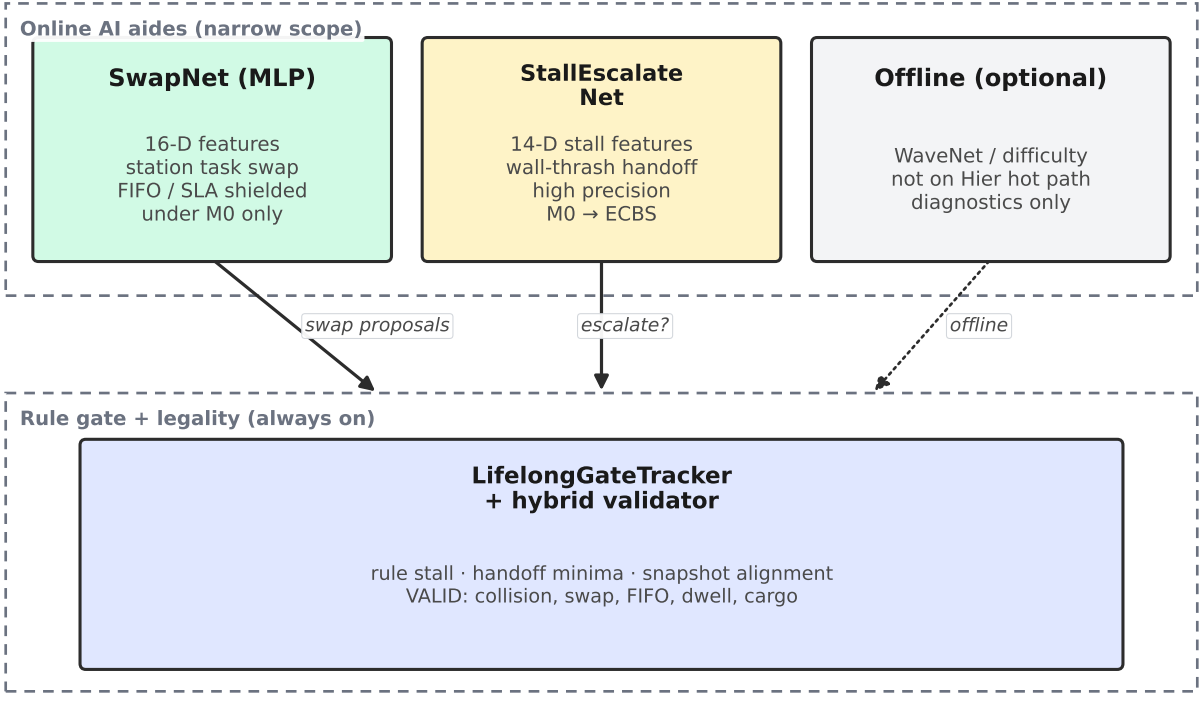AI Modules in the Hierarchical Stack